

any user or operator even knows there is a failure.

Self-healing systems do not just react to events and incidents -- they analyse historic data, identify early triggers or symptoms of failures, and act. For example, if a service is known to crash when it runs out of memory, a self-healing system can observe metrics like memory consumption, predict when the service may fail with very low memory, and take action to fix the issue—like restarting the service or allocating more memory—without human intervention.

In AIOps, self-healing systems are powered by data science in terms of machine learning models, real-time analytics, and automated workflows. The ability for these systems to learn, continue to develop processes to manage, and exist within a controllable design lets organisations work system resiliency into their designs and manage them with minimal manual intervention.

The key shift away from traditional DevOps is that self-healing systems are not just automated -- they are intelligent, autonomous, and adaptive.

Understanding AIOps

AIOps, a term created by Gartner, refers to the automation of problems in traditional IT operations using artificial intelligence technologies such as machine learning and Big Data analytics.

In AIOps, the engine consumes and

ingests a vast number of data points from many sources, including logs, metrics, traces, events, user feedback, etc, and applies intelligent algorithms to:

- Recognise patterns and anomalies
- Predict incidents
- Correlate events, and
- Automatically trigger remediation workflows

AIOps can notify a system administrator of an issue and also act on the insights gained from monitoring events and behaviours in the infrastructure automatically. This could include auto scaling the cloud infrastructure, restarting a failed service, or even reconfiguring systems or environments leading to increased uptime and improved efficiency.

The fusion of AI, ML, and DevOps

At a basic level, AIOps is where DevOps automation intersects with machine intelligence.

- DevOps provides speed, automation, and collaboration across development and operations teams.
- Artificial intelligence (AI) provides pattern matching, decision making and predicting capabilities.
- Machine learning (ML) allows the system to learn from historical incidents and improve over time.

Together, these elements create an intelligent feedback loop. The system processes more data and learns from past behaviour, considers ongoing changing

conditions, and improves its predictions and accuracy based on the data.

What makes a system self-healing?

A self-healing system is a system that can identify faults, determine root causes, and take corrective action without any human intervention as a first response. It models how the human body responds when injured -- identify pain, determine what is wrong, and start the healing process.

Self-healing systems don't just rely on static rules and manual checks; they utilise real-time data streams and apply pattern and anomaly detection through machine learning to ascertain the state of the environment. A self-healing system is trying to gauge its own health all the time — CPU utilisation, latency, memory, throughput, traffic, security anomalies, etc — to preemptively address an impending failure.

The key component of every self-healing system is a cycle that reflects the process followed by intelligent agents: Detect → Diagnose → Act.

Detection: First, the system must be aware that 'something is wrong'. It does this through telemetry data: logs, metrics, traces, user behaviour, and so on, by using anomaly detection algorithms. These telemetry models can detect deviations from normal, like CPU spikes, latency anomalies, and sudden increases in error rates.

Table 1: AIOps vs traditional DevOps: Key distinctions

Aspect	Traditional DevOps	AIOps
Data handling	Manual analysis or rule-based alerting	Real-time ingestion and machine learning-based correlation
Incident response	Human intervention and scripted automation	Automated diagnosis and self-remediation
Scalability	Limited by human capacity and rule complexity	Scales with data and adapts to new patterns
Root cause analysis	Time-consuming and reactive	Fast, proactive, and often predictive
Adaptability	Static logic, predefined workflows	Dynamic learning and evolving models
Alert management	High noise, false positives	Noise reduction and intelligent alert prioritisation